

Pengantar Kecerdasan Artifisial

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CSGE603130 - Kecerdasan Artifisial dan Sains Data Dasar
Semester Genap 2026/2027

Outline

Kecerdasan Artifisial

Definisi Kecerdasan Artifisial

Agen Cerdas

Taksonomi dan Cabang KA

Sejarah Singkat KA

Konsep Dasar Pemelajaran Mesin

KA di Fasilkom UI

Credits

- Stuart Russell & Peter Norvig, "Artificial Intelligence: A Modern Approach", 4th ed., 2020, Chapter 2.
- Ayu Purwarianti, "Teknologi dan Aplikasi Artificial Intelligence", Salindia Modul Pelatihan Thematic Academy, Digital Talent Scholarship, Kemenkominfo 2021.
- Gambar dan tangkapan layar hanya untuk kebutuhan penjelasan
 - Hak cipta tetap ada pada pemilik aslinya.

Kecerdasan Artifisial: *Have you heard of it?*



Program Studi Sarjana

Kecerdasan Artifisial

Dibuka mulai Tahun Ajaran 2026/2027

Jalur Penerimaan

- Seleksi Nasional Berdasarkan Prestasi (**SNBP**)
- Seleksi Nasional Berdasarkan Tes (**SNBT**)
- Prestasi dan Pemerataan Kesempatan Belajar (**PPKB**)
- Seleksi Jalur Prestasi (**SJP**)
- SIMAK UI**

Gelar: Sarjana Komputer (S.Kom.)

Durasi: 4 Tahun (8 Semester)

Persyaratan Daftar:

Lulusan **SMA/ sederajat** dengan latar belakang akademik yang kuat, terutama pada mata pelajaran **Matematika** dan **IPA**, sebagai dasar dalam mengikuti proses pembelajaran.

Karir:

- AI scientist
- Machine learning scientist
- AI developer
- AI Ops engineer atau MLOps engineer
- Big data analyst
- Data engineer
- Knowledge engineer

Cek selengkapnya 

<https://cs.ui.ac.id>

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Like Reply

 **...**

Mana kelas prompting-nya 💔

36 likes Reply

— View all 1 replies

 **...** 1w

Ini diajarin cara nge-prompt ya?

1 like Reply

 **syarna_** 1w

Lah gabeda jauh sama cs trnyt

1 like Reply

— View all 1 replies

 **ikky_code.tech** 1w

keren, tapi takut kedepannya udh kebanyakan anak it.. mending tani aia lh

3K 27

January 13

Add a comment...

ChatGPT 5.2 ▾

↑ Share ...

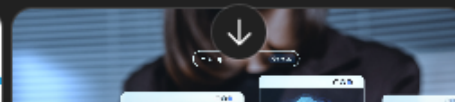
Do people know about AI?

Short answer: **yes—but unevenly and often shallowly.**

Most people *know of* AI, but fewer really *understand* it. Here's the landscape in plain terms:

🌐 General public

- High awareness, low depth
- People associate AI with things like:
 - ChatGPT / chatbots
 - Face recognition
 - Recommendation systems (Netflix, Shopee, TikTok)
- But many still think AI is either:
 - 🧠 *almost human* (too magical), or
 - 🤖 *just automation* (too trivial)



+ Ask anything

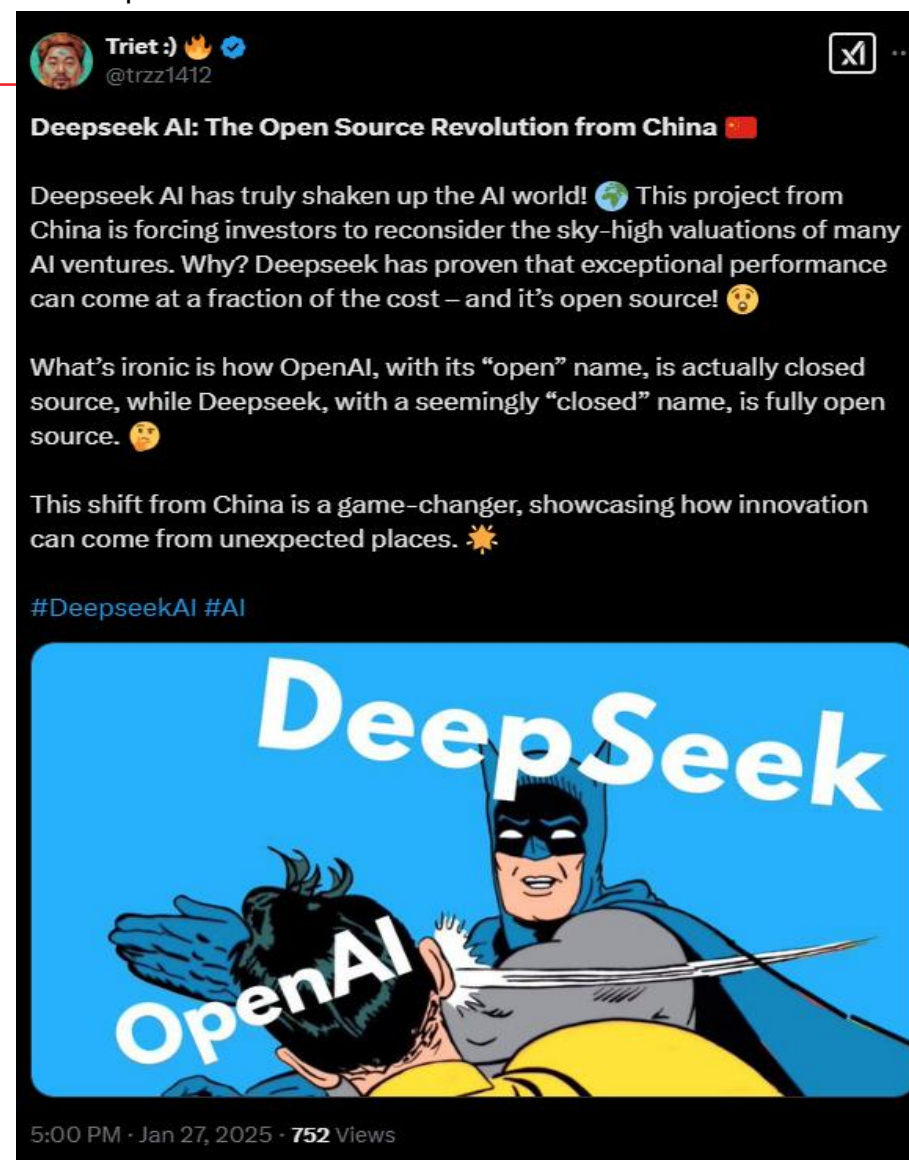


ChatGPT can make mistakes. Check important info. See [Cookie Preferences](#).

<https://www.instagram.com/ecommurz/p/DFXfwXfvWcv>



<https://x.com/trzz1412/status/1883817431145308509>





- Deepseek was built by super smart guys with **top math background**
- Don't knock math~ It's quite important
- Nowadays a lot of *black-box solutions* are out there – **using** them does not make you an AI expert.

<https://x.com/hxiao/status/1882099204044239007>

Global Stance on AI



2021 AI Index Report Ch. 7 (<https://aiindex.stanford.edu>)

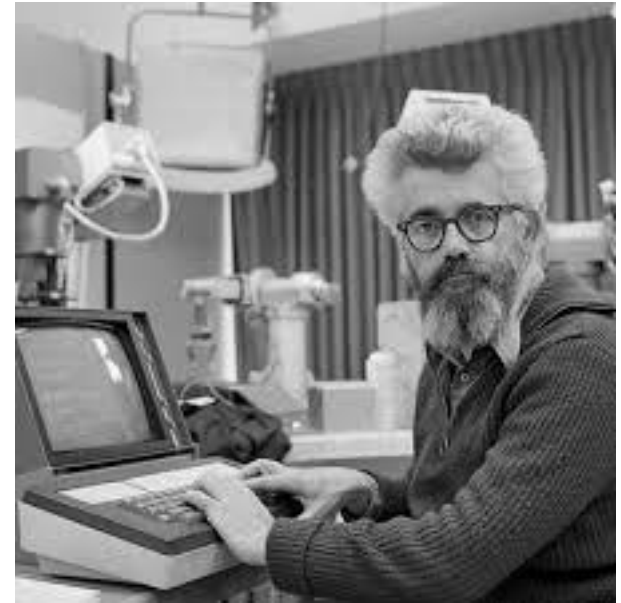
Kecerdasan Artifisial di Indonesia



Apa itu Kecerdasan Artifisial?

What is AI?

- **Q. What is artificial intelligence?**
- A. It is the science and engineering of **making intelligent machines**, especially intelligent computer programs.
- **Q. Yes, but what is intelligence?**
- A. Intelligence is the **computational part of the ability to achieve goals** in the world, performed by people, many animals and some machines.
- **Q. Isn't there a solid definition of intelligence that doesn't depend on relating it to human intelligence?**
- A. **Not yet.** The problem is that we cannot yet characterize in general what kinds of computational procedures we want to call intelligent.



What is AI? – Thought vs. Behavior, Human vs. Rationality

Thinking **humanly**

"The exciting new effort to make computers think ... machines with minds, in the full and literal sense."
(Haugeland, 1985)

"[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning ..." (Bellman, 1978)

Thinking **rationally**

"The study of mental faculties through the use of computational models." (Charniak and McDermott, 1985)

"The study of the computations that make it possible to perceive, reason, and act." (Winston, 1992)

Acting **humanly**

"The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil, 1990)

"The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991)

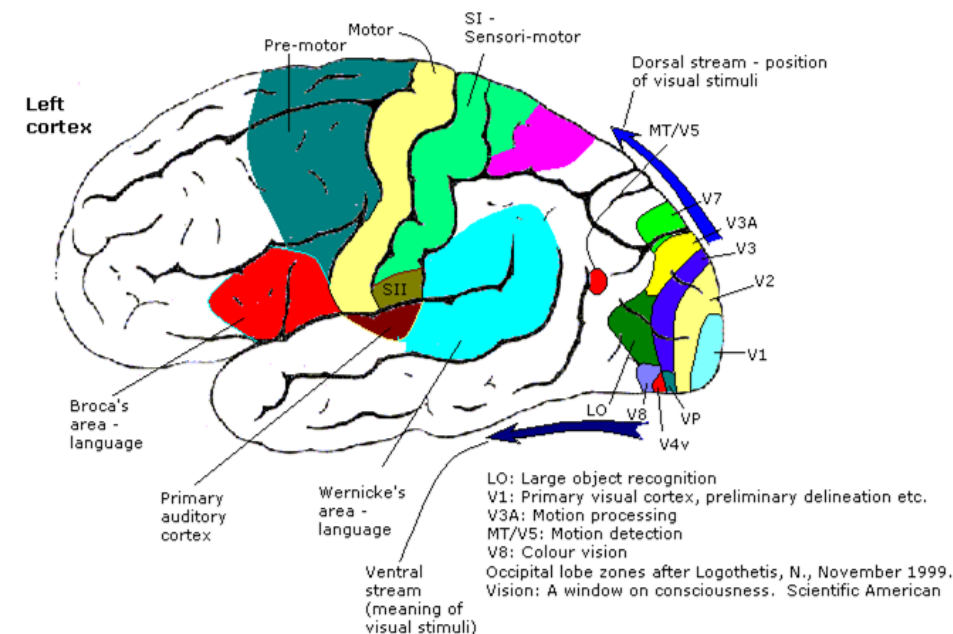
Acting **rationally**

"Computational Intelligence is the study of the design of intelligent agents." (Poole et al., 1998)

"AI ...is concerned with intelligent behavior in artifacts."
(Nilsson, 1998)

Thinking Humanly: Cognitive modeling

- Menjawab "apakah mesin berpikir seperti manusia?"
perlu jawaban dari "bagaimana manusia berpikir?"
 - Introspeksi: "*catch our own thoughts as they go by*"
 - Eksperimen psikologi: observasi perilaku manusia
 - *Brain imaging*: observasi otak manusia.
- *Cognitive science*: model komputasi dari KA + psikologi eksperimental -> teori akal manusia.
 - Contoh: model *neuroimaging* → model komputasi *convolutional neural networks*.
- **Kritik** (di awal) – kebingungan: model KA oke, maka teori akal oke; atau, sebaliknya?
 - Sekarang, kedua klaim dipisah



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Acting humanly: Turing Test

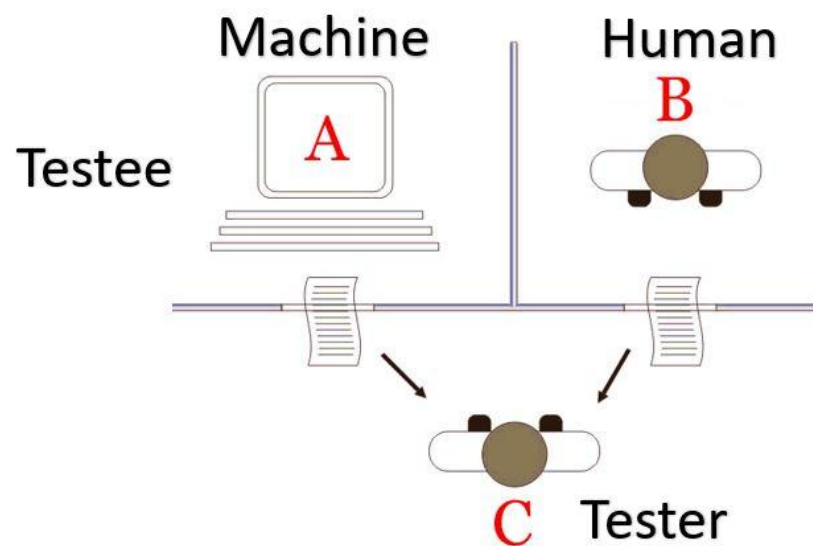


Image by Juan Alberto Sánchez Margallo, CC BY 2.5, via
Wikimedia Commons

- Thought experiment to avoid question "Can a machine think?"
- **C** (tester) bertanya ke A dan B beberapa pertanyaan.
- **A** berusaha mengelabui C hingga C mengira A manusia.
- **A** harus menampilkan perilaku cerdas, menyamai performa manusia dalam tugas-tugas kognitif:
 - Pemrosesan bahasa (komunikasi)
 - Representasi pengetahuan
 - Penalaran otomatis
 - Adaptasi ke situasi baru (machine learning)
- Pada total Turing test, juga perlu:
 - Computer vision (persepsi objek)
 - Robotika (*act upon objects*)
- **Kritik:** Pesawat terbang sukses tidak dengan meniru bagaimana burung terbang. Jadi sukses di KA tidak dengan meniru bagaimana manusia bertindak.

Thinking Rationally: "Laws of Thought"

- Rasionalitas: "*the right thinking*"
 - Penalaran yang tak terbantahkan
- Wujud: pendekatan **logika**
 - *Classical **propositional** and **first-order logic***
 - Bahasa logika probabilistik untuk menangani ketidakpastian
 - Secara teori, menghasilkan model rasionalitas mencakup: informasi persepsi mentah → pemahaman tentang dunia → prediksi masa depan
- **Kritik**: TIDAK menghasilkan perilaku cerdas
 - Manusia tidak selalu rasional
 - Tak semua hal dapat diekspresikan dengan logika
 - Tidak *scalable*.

$$\begin{aligned} & ((p \rightarrow q) \wedge p) \vdash q \\ & ((p \rightarrow q) \wedge \neg q) \vdash \neg p \\ & ((p \rightarrow q) \wedge (q \rightarrow r)) \vdash (p \rightarrow r) \\ & ((p \vee q) \wedge \neg p) \vdash q \end{aligned}$$

Acting rationally: Pendekatan agen rasional

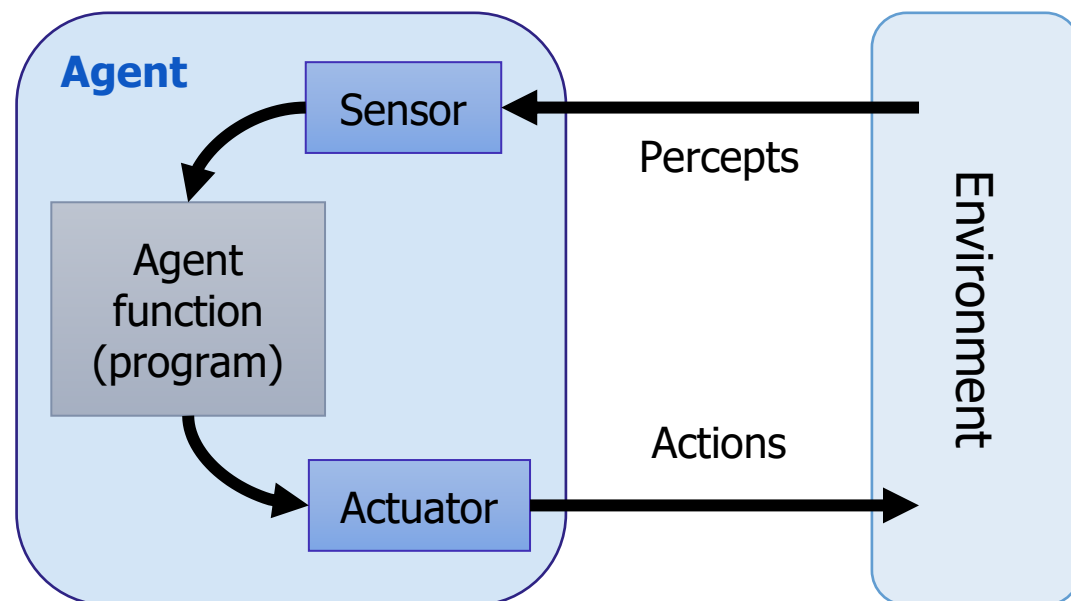
- **“Do the right thing”** – Bertindak untuk mencapai hasil (yang diharapkan) terbaik.
- Mencakup "thinking rationally", tapi juga perilaku yang muncul tanpa penalaran (mis: refleks jika menyenggol kompor panas)
- Lebih dapat dimodelkan secara matematis (dibandingkan yang lain) sehingga dapat dikembangkan secara ilmiah dan menjadi **model standar KA**.
- The right thing dinyatakan dengan tujuan yang disuplai ke agen.
 - meminimalkan fungsi biaya, memaksimalkan reward /utility
- Perfect rationality (selalu mengambil tindakan yang terbaik) bisa tidak feasible di lingkungan kompleks.
 - Perlu memutuskan tindakan terbaik dalam batasan/constraint komputasi tertentu



<https://www.flickr.com/photos/nationalmuseumofamericanhistory/8682679829>

Agen Cerdas

Agent Design



- Percept $\mathcal{P}$ adalah konten persepsi melalui sensor
- Percept sequence $\mathcal{P}^*$ adalah *complete history* berisi seluruh hasil persepsi yang diperoleh agen
- *Agent function* dapat dinyatakan secara matematis sebagai $f: \mathcal{P}^* \rightarrow \mathcal{A}$ yang memetakan percept sequence $\mathcal{P}^*$ ke *action* $\mathcal{A}$
- *Agent program* adalah implementasi konkrit fungsi f pada sistem tertentu

Rasional = "Do the right thing"????



Photo by [YoonJae Baik](#) on [Unsplash](#)

- Dari mana tahu agen sudah "do the right thing"?
- Gunakan **ukuran kinerja** (*performance measure*) pada keadaan lingkungan.
- Fokus pada efek lingkungan yang diinginkan, bukan pada *action* agen.
 - Apa akibatnya jika kinerja robot *vacuum cleaner* dalam jumlah debu yang dibersihkan dalam 8 jam?
- Rasionalitas → *expected performance*.

Lingkup Tugas *Agent*

- Bergantung pada tugas/*task* yang akan dilakukan agen.
- Deskripsi lingkungan = PEAS (contoh pengemudi taksi):
 - **Performance** = ukuran kinerja, mis: aman, cepat, tidak melanggar rambu lalin, nyaman bagi penumpang
 - **Environment** = lingkungan, mis: jalan, rambu lalin, kendaraan lain, penumpang
 - **Actuators**, mis: kemudi, gas, rem, klakson
 - **Sensors**, mis: kamera, sonar, speedometer, GPS

Environment Types

Fully observable (vs. partially observable)	Sensor memberikan akses seluruh lingkungan secara lengkap.
Single-agent (vs. multiagent)	Agen bekerja sendiri (tanpa kehadiran agen lain) di lingkungan; bisa bersifat kompetitif, atau kooperatif.
Deterministic (vs. nondeterministic)	Keadaan lingkungan selanjutnya dapat ditentukan sepenuhnya dengan pasti hanya berdasarkan keadaan lingkungan sekarang dan tindakan yang dilakukan agen. Disebut stokastik jika ada nilai peluang yang dinyatakan secara eksplisit.
Episodic (vs. sequential)	Pengalaman agen terbagi ke dalam episode-episode atomik yang masing-masing terdiri dari sepasang <i>percept</i> dan <i>action</i> , dan pilihan <i>action</i> di satu episode tidak berpengaruh pada pilihan <i>action</i> di episode yang lain.
Static (vs. dynamic)	Lingkungan tidak berubah pada saat agen sedang dalam "berpikir". Semi-dynamic: lingkungan tak berubah, tapi skor kinerja dapat berubah.
Discrete (vs. continuous)	Berlaku pada banyaknya <i>state</i> , kemungkinan <i>percepts</i> , pilihan <i>actions</i> , dan bagaimana variabel waktu dimodelkan.
Known (vs. unknown)*	Lebih pada pengetahuan agen (atau perancangnya) tentang "hukum fisik" dari lingkungan. Known: semua <i>outcome</i> atau <i>outcome probabilities</i> dapat diketahui dari setiap kemungkinan <i>action</i> .

Environment Examples

	Chess with a clock	Chess without a clock	Taxi driving
Fully observable?	Yes.	Yes.	No (partial).
Single agent?	No.	No.	No.
Deterministic?	Yes.	Yes	No.
Episodic?	No.	No.	No.
Static or (semi) dynamic?	Semi-dynamic.	Yes.	No.
Discrete?	Yes.	Yes.	No.

- Sifat "known/unknown" bukan sepenuhnya sifat lingkungan sendiri. Pemain catur bekerja pada *known environment* karena semua aturan lingkungan diketahui.
- *Known environment* dapat bersifat *partially observable* (contoh: permainan Solitaire).
- *Unknown environment* dapat bersifat *fully observable* (contoh: pada *video game* baru, layar menunjukkan seluruh keadaan lingkungan, tapi kita belum tahu apa yang terjadi ketika tombol *controller* ditekan).

Environment Exercise

	<Fill something here>
Fully observable?	
Single agent?	
Deterministic?	
Episodic?	
Static or (semi-)dynamic?	
Discrete?	

- Dunia nyata (*real world*) bersifat *partially observable*, *multi-agent*, *non-deterministic*, *sequential*, *dynamic*, *continuous*, dan *unknown*.

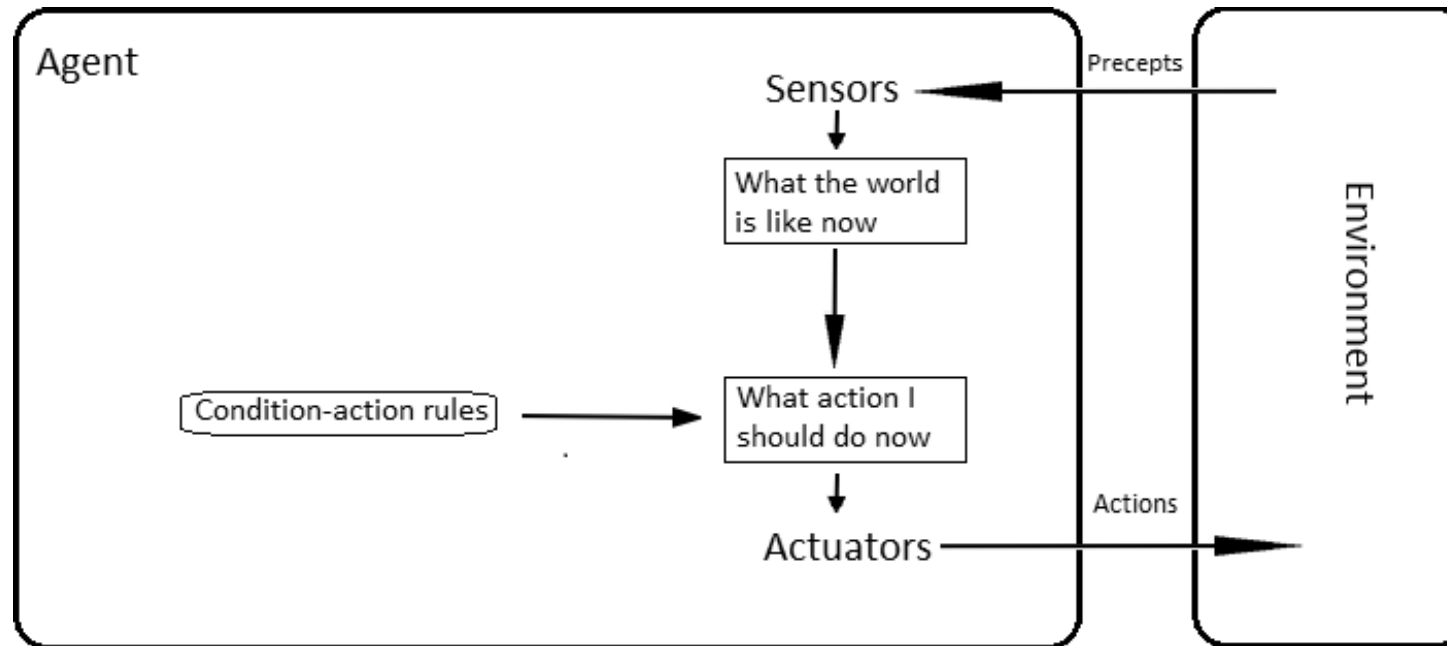
Type-tipe Dasar Program Agent

- Simple reflex agent
- Model-based reflex agent
- Goal-based agent
- Utility-based agent



Bisa diubah menjadi
Learning Agent

Simple Reflex Agent



Contoh condition-action-rule:
if car-in-front-is-braking, then initiate braking

Image source: DDSniper, CC0, via Wikimedia Commons

- Perilaku refleks sederhana
- Kecerdasan terbatas
- Bekerja baik hanya jika lingkungan fully observable.

Model-based Reflex Agent

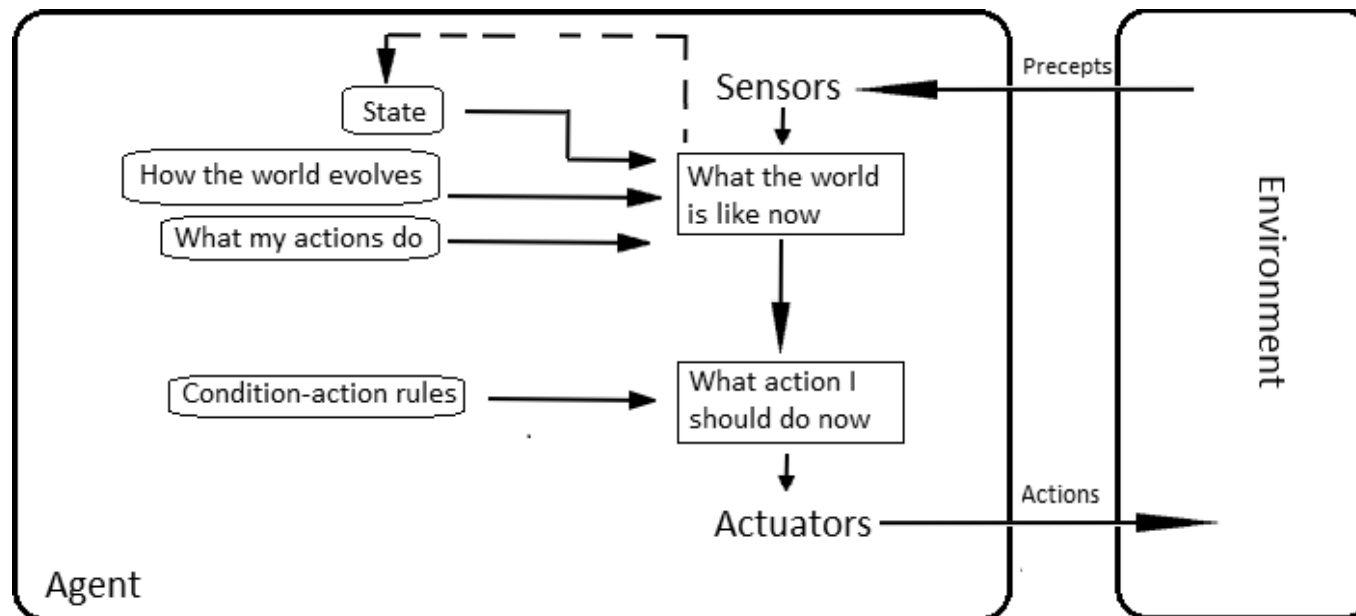
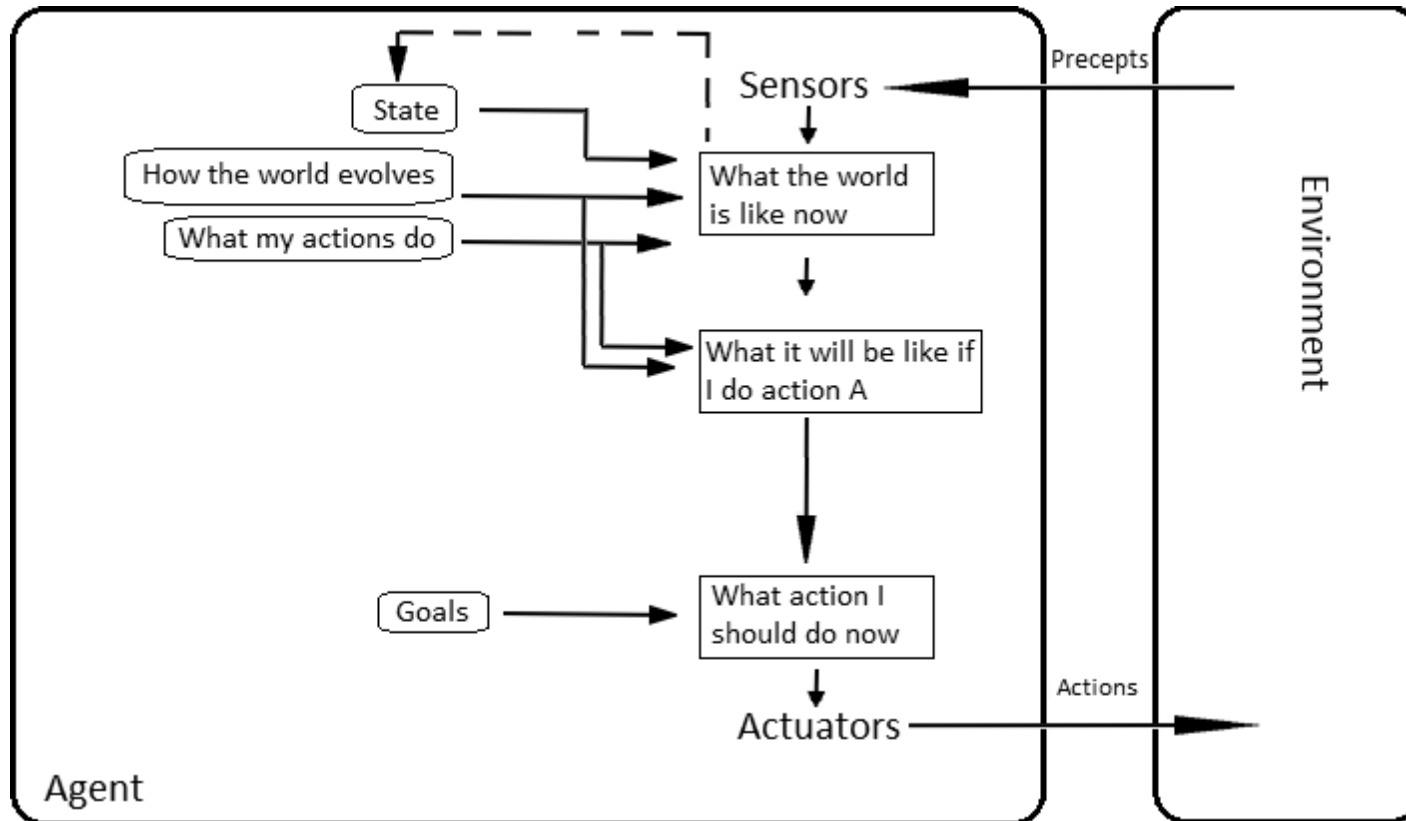


Image source: DDSniper, CC0, via Wikimedia Commons

- "Putar setir ke kanan" → "mobil belok kanan"
- "Kamera basah" → "Droplet di citra yang tertangkap"

- State internal (sederhana/kompleks) merekam (sebagian) keadaan lingkungan melalui:
 - model transisi lingkungan (efek *action* atau perubahan lingkungan independen)
 - model sensor (refleksi keadaan lingkungan yang tertangkap sebagai *percept*)
- "What the world is like now" → "best guess(es)" di lingkungan yang *partially observable*.

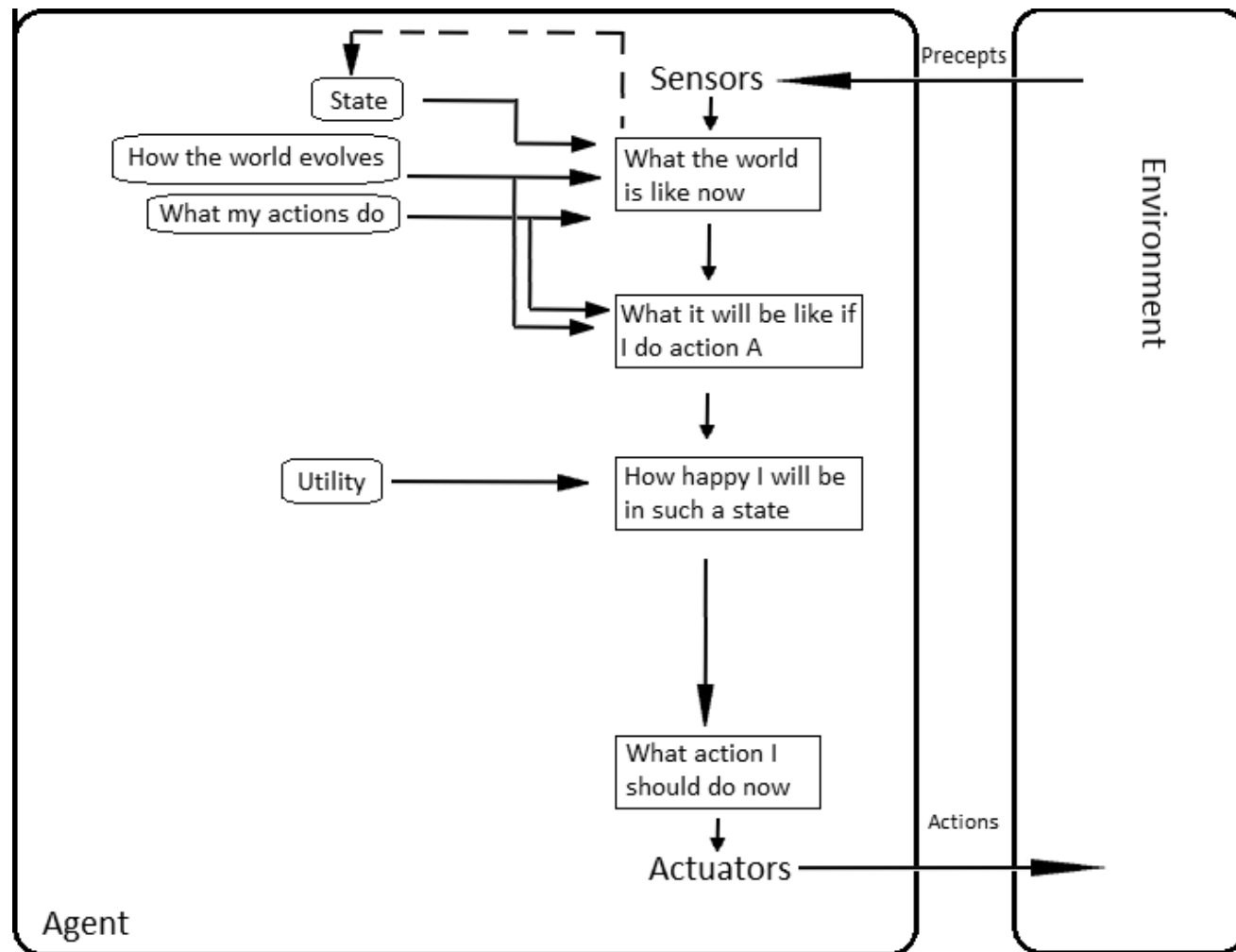
Goal-based Agent



Source: DDSniper, CC0, via Wikimedia Commons

- Model + Informasi *goal* yang diinginkan.
- Pemilihan *action* mempertimbangkan *goal* (masa depan)
- Pencarian sekuens *action* menuju *goal* melalui **search** atau **planning**

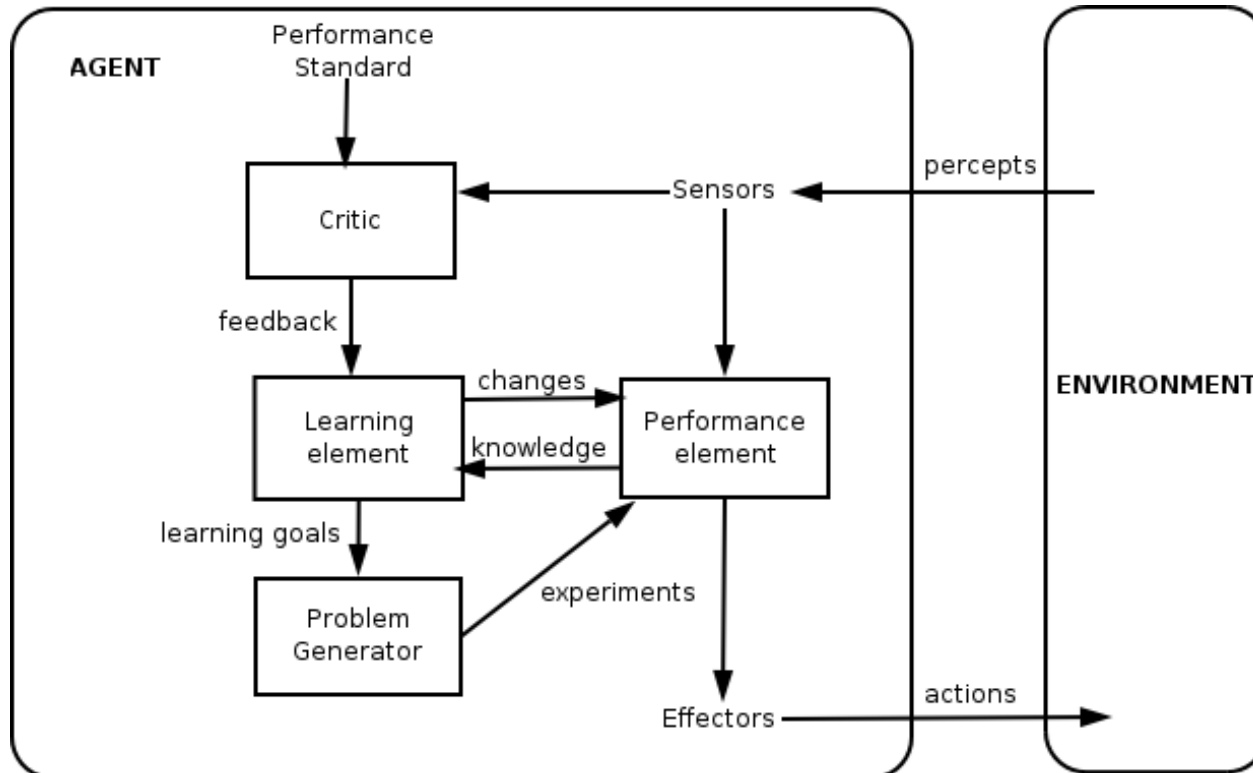
Utility-based Agent



Source: DDSniper, CC0, via Wikimedia Commons

- Banyak cara menuju *goal* → pilih yang paling disukai
 - Lebih cepat, lebih aman, lebih murah, dll.
- *Performance measure* mengukur keadaan lingkungan yang paling disukai.
- *Utility function*: internalisasi ukuran kinerja dari perspektif agen.
- Pilih *action* yang memaksimalkan *expected utility*.
- Bisa menangani lingkungan yang *partially observable* dan nondeterministik

Learning Agent



Source: Utkarshraj Atmaram, Public domain, via Wikimedia Commons

- Learning agent lebih pada bagaimana agen terwujud, bukan pada cara memilih *action*.
- Semua (model-based, goal-based, utility-based) agen dapat dirancang sebagai *learning agent*.
- Komponen:
 - *Performance element*: bertugas memilih *action*
 - *Learning element*: bertugas memperbaiki *state* internal agen dengan bantuan *critic*.
 - *Problem generator*: menyarankan *action* eksploratif agar tidak "itu-itu" saja.

Sejarah Singkat KA



The history of AI

1940s-1950s

Foundations of AI

In the 1940s, the first artificial neurons were conceptualised. The 1950s introduced us to the Turing Test and the term "Artificial Intelligence".



1960s-1970s

Early Development

The 60s and 70s brought the birth of ELIZA, simulating human conversation, and Dendral, the first expert system, showcasing the early potentials of AI.



1980s

AI Winter & Expert Systems

The 80s faced reduced AI funding but saw the inaugural National Conference on AI. The backpropagation concept rejuvenated neural networks.



1990s

Revival & Emergence of ML

The 90s witnessed IBM's Deep Blue defeating chess champion Garry Kasparov and the inception of the LOOM project, laying the foundations for GenAI.



2000s

The Genesis of Generative AI

Geoffrey Hinton propelled deep learning into the limelight, steering AI toward relentless growth and innovation.



2010s

Rise of AI

In 2011, IBM Watson won "Jeopardy!", highlighting AI's language skills. The 2010s marked major AI milestones, including pioneering work in image recognition and the birth of GANs in 2014, followed by OpenAI's founding in 2015.



2020s

GenAI Reaches New Horizons

At the start of this decade, we've seen significant strides in GenAI, notably with OpenAI's GPT-3 and DALL-E. 2023 welcomed advanced tools like ChatGPT-4 and Google's Bard, alongside Microsoft's Bing AI, enhancing accessibility and reliability of information.



<https://www.calls9.com/blogs/the-history-of-ai-a-timeline-from-1940-to-2023>

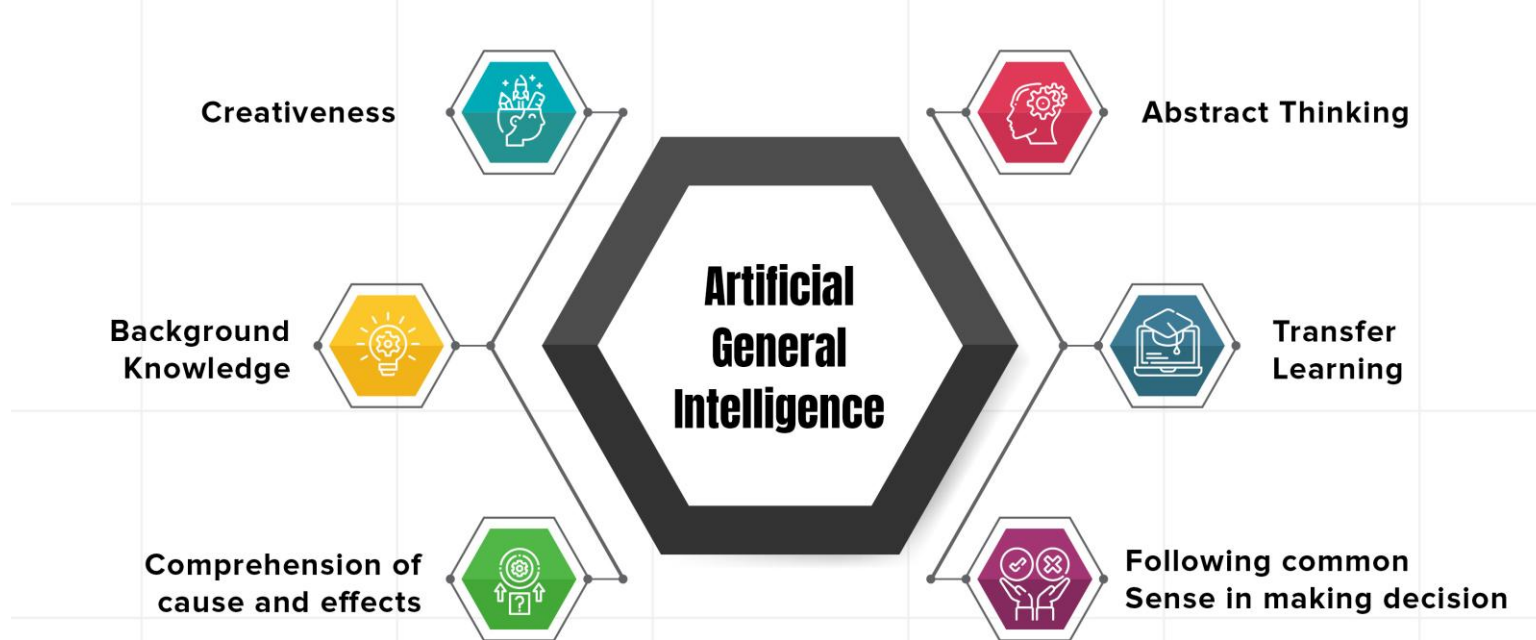
Taksonomi KA

KA berdasarkan Lingkup Tugas (*Task Scope*)

	Artificial (Narrow) Intelligence – Weak AI	Artificial General Intelligence – Strong AI
Ide	Menangani hanya satu tugas tertentu saja. Kecerdasan hanya terbatas pada tugas tersebut, dan tidak mampu melakukan tugas-tugas kompleks seperti kecerdasan manusia.	Dapat menangani semua tugas kompleks yang dilakukan manusia. Mampu menyamai (mungkin melebihi?) kecerdasan manusia.
Area penerapan	Bekerja baik dalam kasus yang mana mesin harus melakukan tugas-tugas berulang secara otomatis.	Mesin harus mampu berpikir, melakukan penalaran, dan semua fungsi yang dapat dilakukan manusia
Sifat tugas	Berulang/repetitive, tidak melibatkan pengambilan keputusan secara otonom	Melakukan pengambilan keputusan dan analisis secara otonom
Contoh	Chatbot reservasi restoran, sistem klasifikasi aplikasi kredit di bank,	??? (belum ada; masih dalam riset)

Artificial General Intelligence (AGI)

- AGI is a type of intelligence that is flexible and general that can outperform humans at most (if not all) intellectual tasks.
- ChatGPT is as an early yet still incomplete version of AGI.

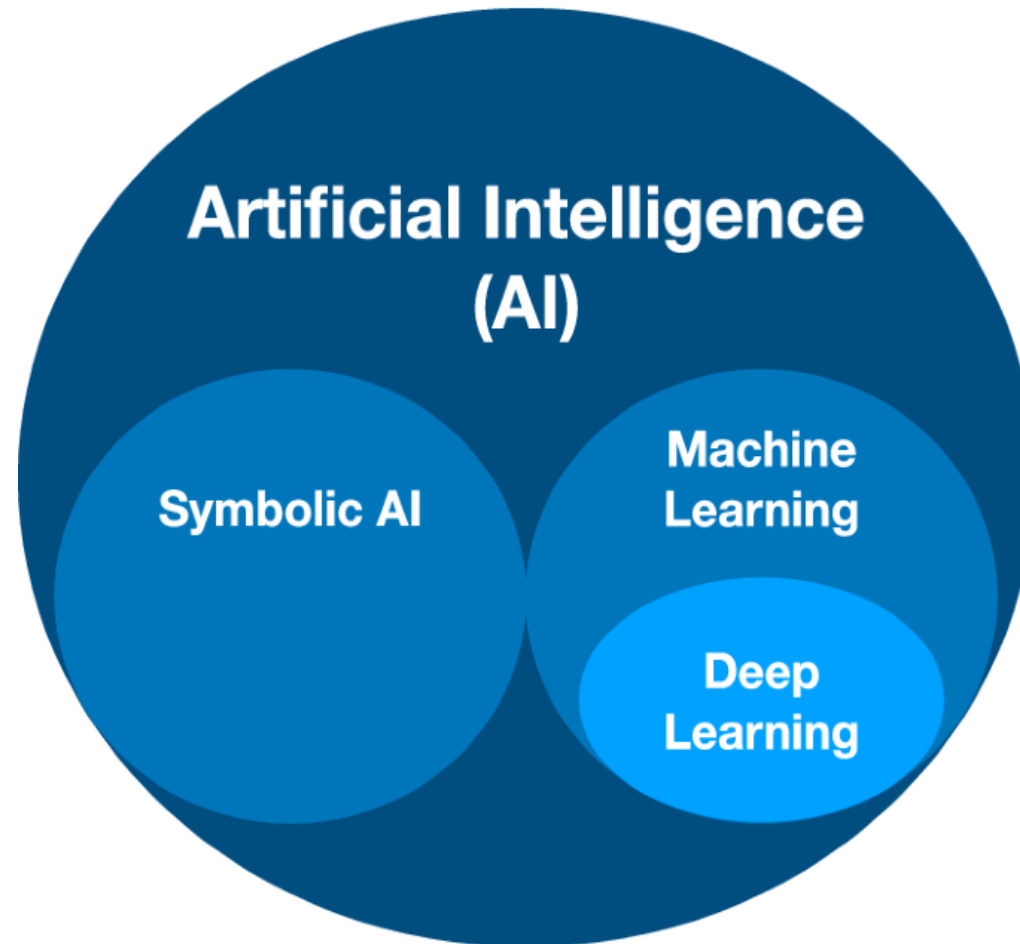


KA berdasarkan ranah/domain permasalahan

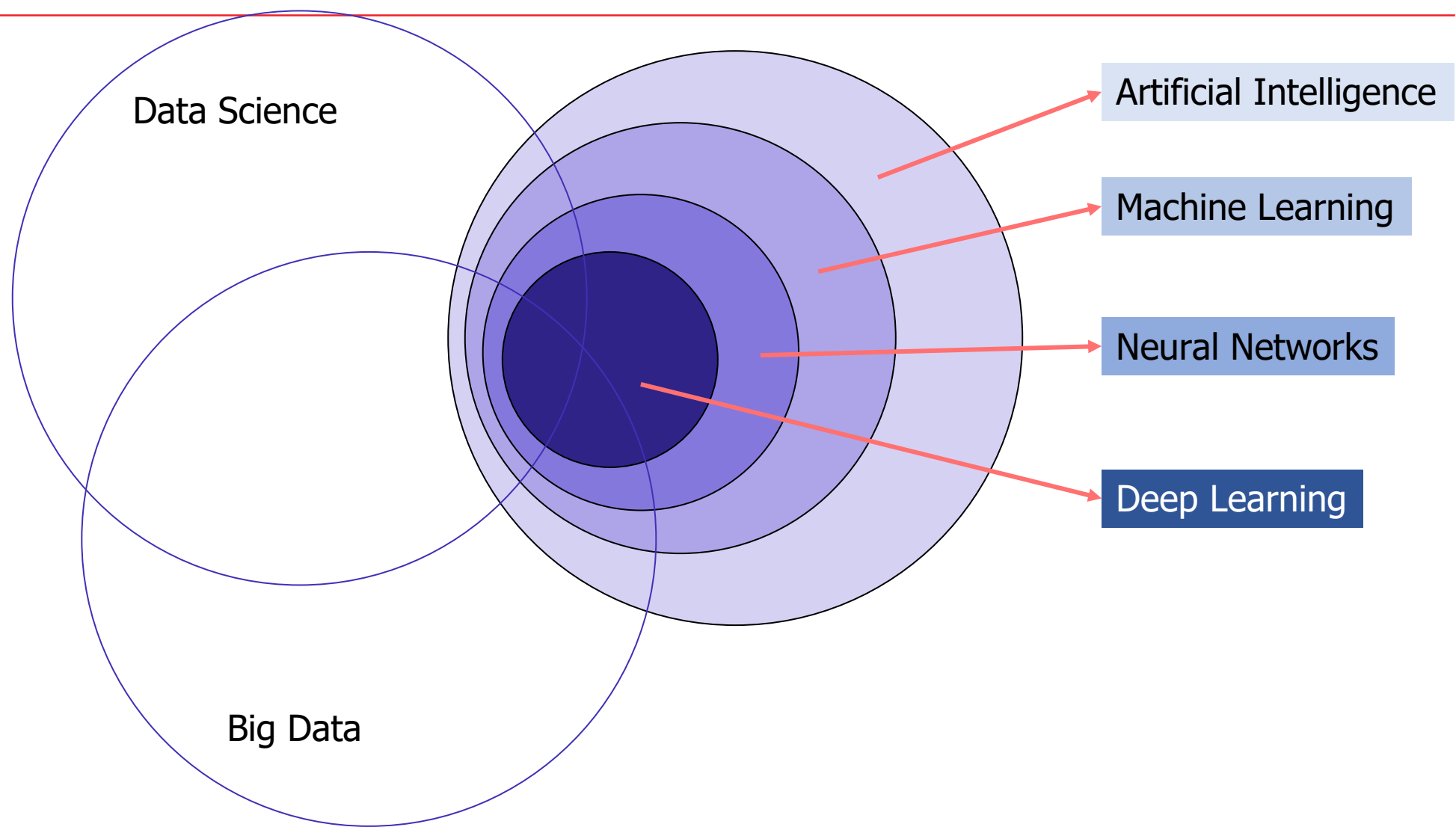
- Contoh subbidang KA berdasarkan ranah permasalahan:
 - natural language processing – fokus pada teks bahasa,
 - speech processing – fokus pada sinyal bicara
 - image processing/computer vision – fokus pada citra/video,
 - knowledge representation and reasoning – fokus pada representasi pengetahuan berbasis bahasa logika formal
 - dan lain-lain ...
- Banyak ranah permasalahan KA yang membutuhkan paduan dari beberapa subbidang tersebut.
 - pengenalan bahasa isyarat – dari citra/video ke teks bahasa, dan sebaliknya

AI Taxonomy

- You can always find someone to debate you on this

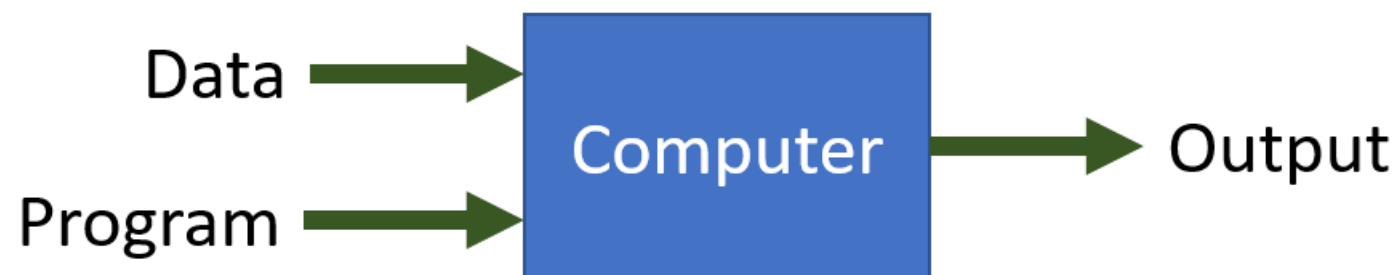


AI, Big Data and Data Science

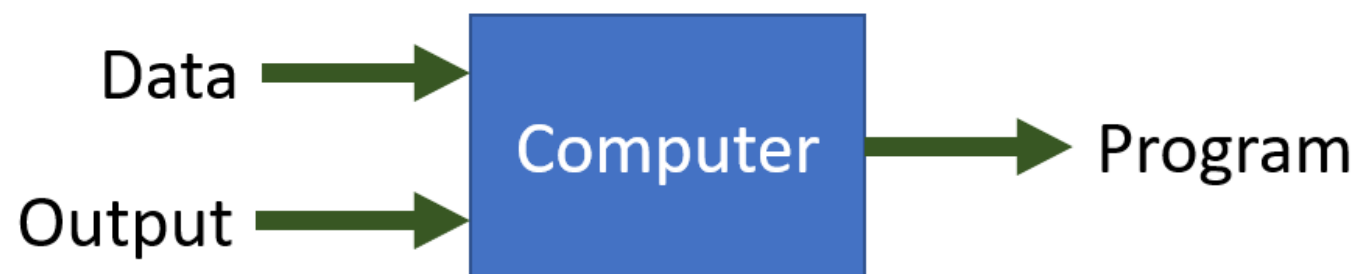


Konsep Dasar Pemelajaran Mesin

Traditional Programming vs. Machine Learning

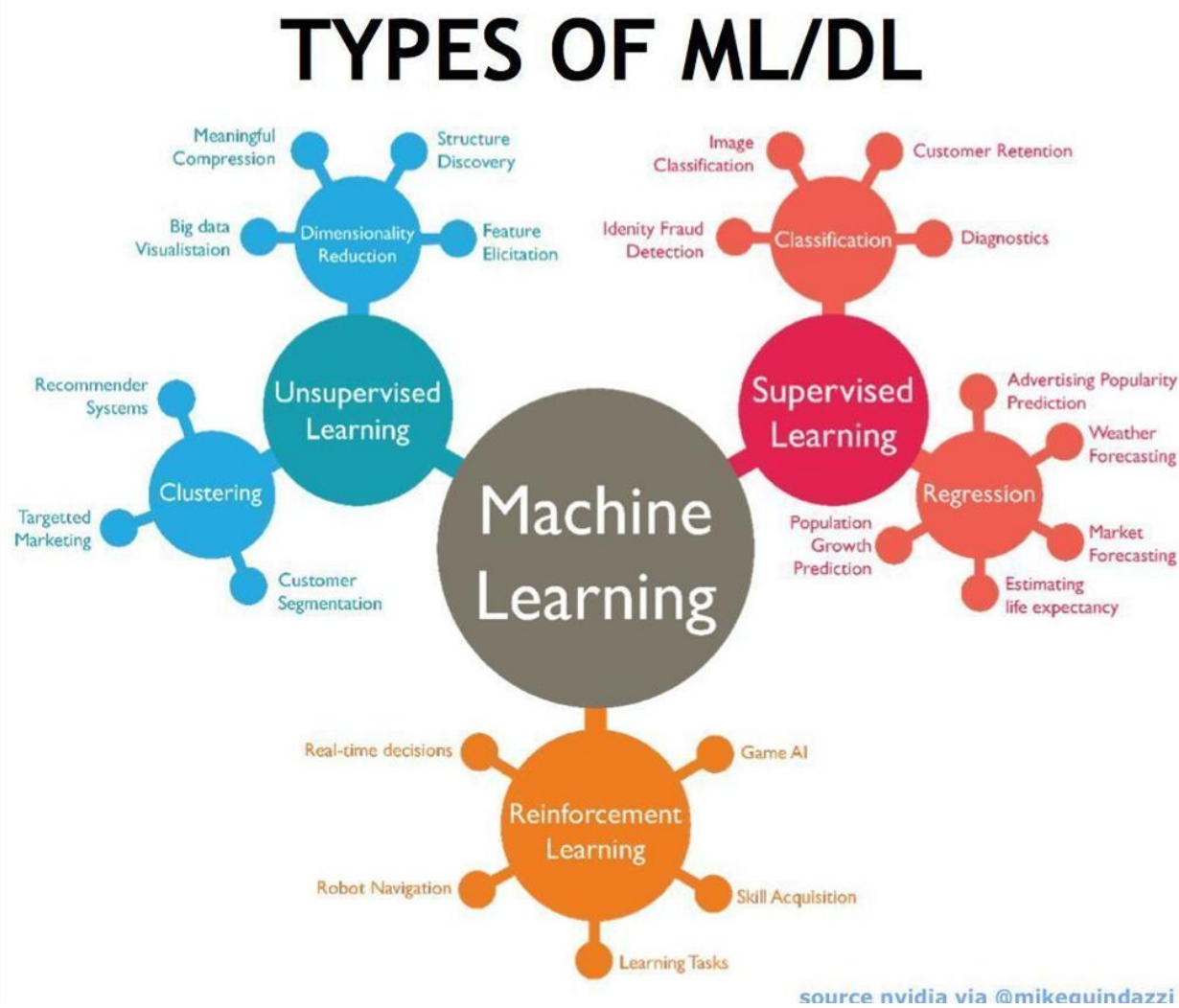


Traditional programming

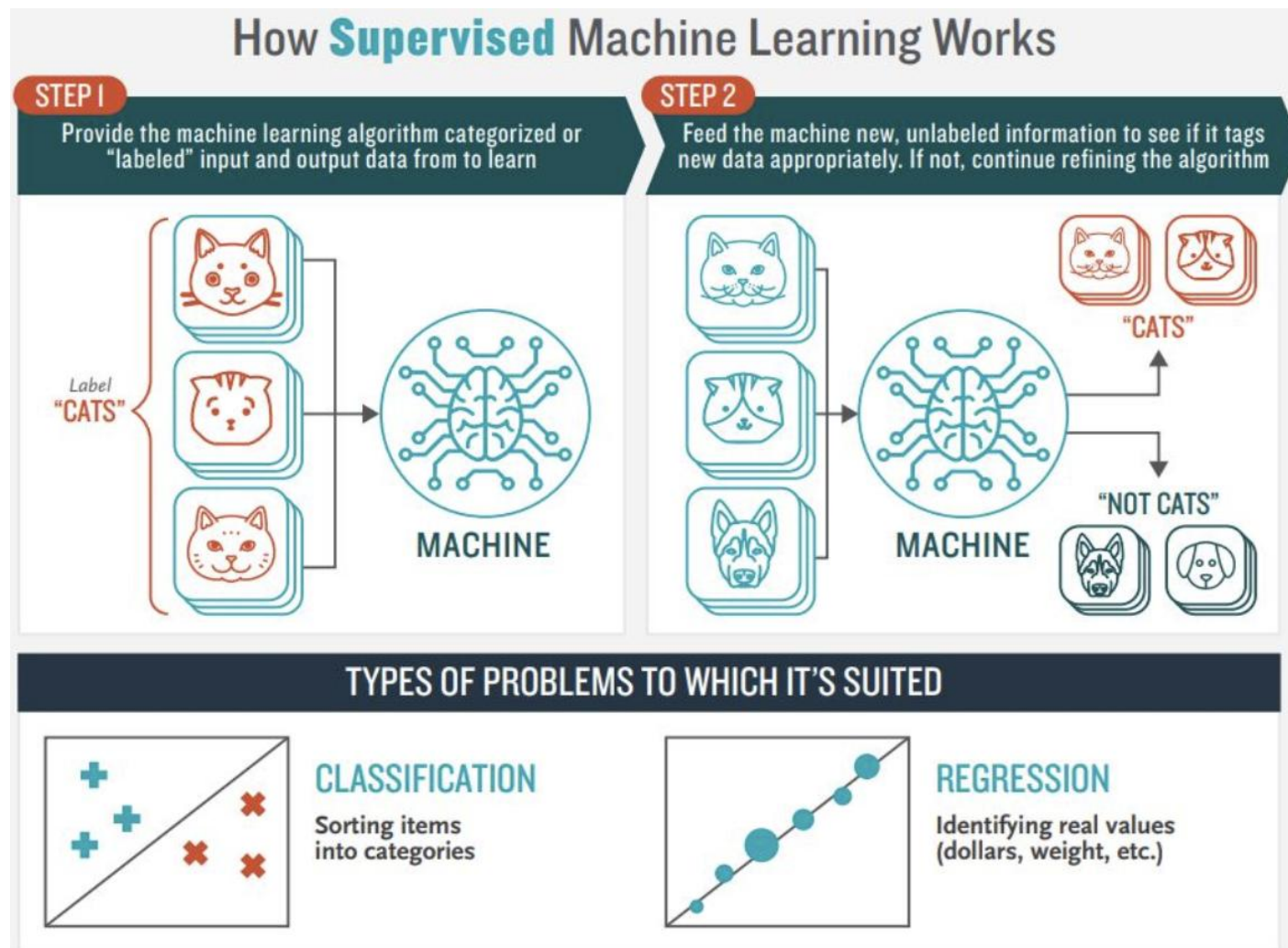


Machine Learning

Types of Machine Learning



Supervised (Inductive) learning



- Diberikan data training, contoh respons yang benar diberikan ke algoritma.
- Algoritmanya melakukan generalisasi sehingga dapat memberikan respons yang benar untuk semua kemungkinan input (termasuk yang tidak termasuk dalam data training).

<https://www.lotus-qa.com/data-annotation-guide/>
<https://medium.com/@jorgesleonel/supervised-learning-c16823b00c13>

Classification

Contoh: memprediksi seseorang beli komputer atau tidak

- **Data:** Himpunan records (atau samples, atau instances) yang terdiri dari:
 - **attributes:** $A_1, A_2, \dots, A_k$.
 - **class:** setiap sample diberi label dengan kelas tertentu
- **Goal:** mempelajari **classification model** dari data yang dapat digunakan untuk memprediksi kelas dari instances baru yang belum diketahui kelasnya.
- Prediksi bernilai **discrete**.

Age	income	Student?	Credit rating	Buys computer?
< = 30	high	no	fair	no
< = 30	high	no	excellent	no
31...40	high	no	fair	yes
> 40	medium	no	fair	yes
> 40	low	yes	fair	yes
> 40	low	yes	excellent	no
31...40	low	yes	excellent	yes
< = 30	medium	no	fair	no
< = 30	low	yes	fair	yes
> 40	medium	yes	fair	yes
< = 30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
> 40	medium	no	excellent	no

Regression

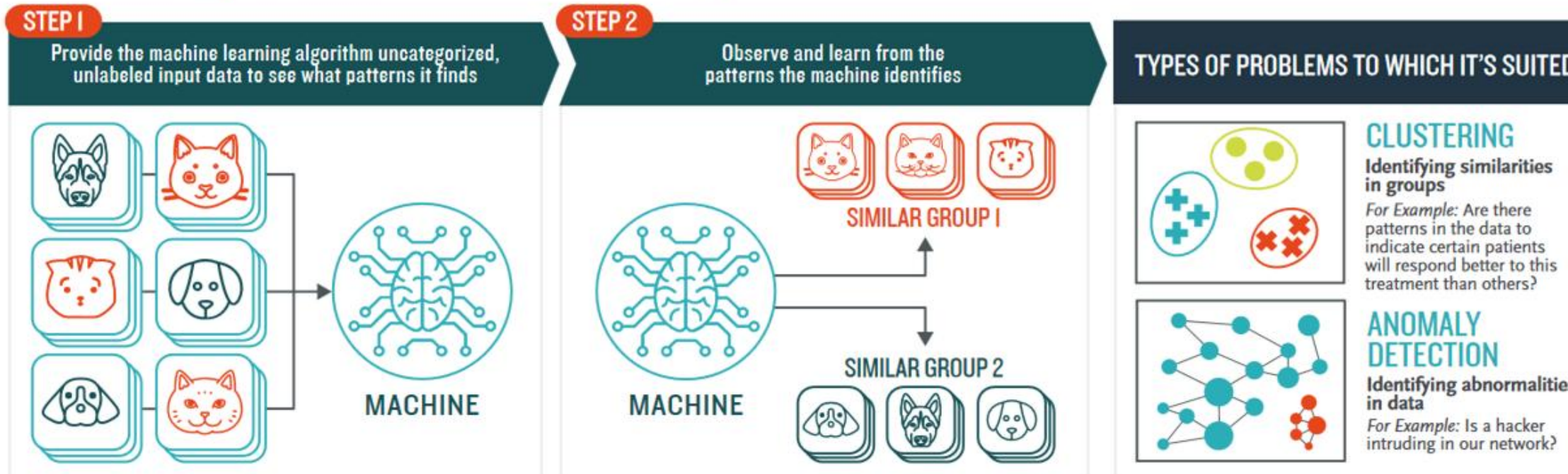
Contoh: Memprediksi Harga Mobil Bekas

- Klasifikasi** mencoba memprediksi **kelas** dari suatu instance data, sedangkan **regresi** mencoba memprediksi suatu **nilai (bilangan continues)**.

brand	model	year	miles	Horse power	engine torque	engine capacity	fuel capacity	num_cylinder	num_seat	num_owner	price
Chevrolet	Volt LT	2019	27173	149	294	15	89	4	5	3	25655
	Elantra										
Hyundai	SE	2017	76941	146	132	2	14	4	5	2	15590
Ford	Focus	2014	97027	159	146	2	124	4	5	3	13590
Honda	Civic	2016	95396	158	138	15	124	4	5	2	20590
	Elantra										
Hyundai	SE	2019	28405	147	132	2	14	4	5	3	16787
Honda	Civic	2016	61459	158	138	2	124	4	5	2	18277
Subaru	Legacy	2022	6811	182	176	25	185	4	5	1	33777
Honda	Clarity	2018	29674	212	99	15	7	4	5	1	26999
Volkswagen	Jetta	2019	25044	147	184	14	132	4	5	1	22491
Volkswagen	Jetta	2017	26215	150	184	14	145	4	5	2	17500
Honda	Civic	2015	25939	143	129	18	132	4	5	1	19450
Honda	Civic	2019	32270	174	162	15	124	4	5	1	22688
Honda	Civic	2018	19950	158	138	2	124	4	5	1	22988
Honda	Accord	2020	19719	192	192	15	148	4	5	1	26988
Honda	Civic	2020	16076	174	162	15	124	4	5	1	24498
Subaru	Impreza	2019	28214	152	145	2	132	4	5	1	22360

Unsupervised Learning

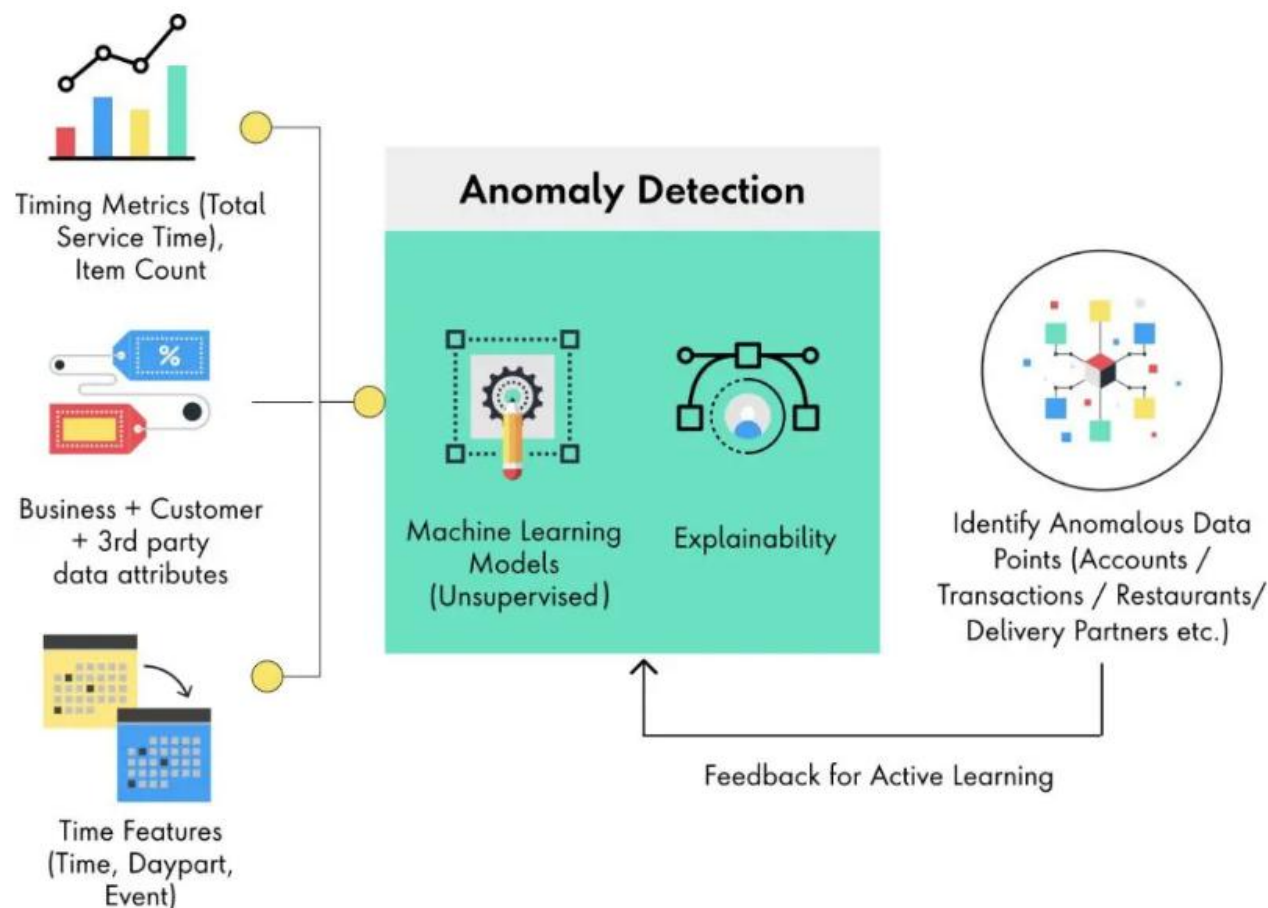
How **Unsupervised** Machine Learning Works



- Diberikan data training, contoh respons yang benar tidak diberikan ke algoritma.
- Algoritmanya mengidentifikasi kemiripan antar input sehingga yang mirip akan dikelompokkan (clustered) bersama.

<https://www.lotus-qa.com/data-annotation-guide/>

Clustering: Fraud Detection



Anomaly Detection for Potential Frauds

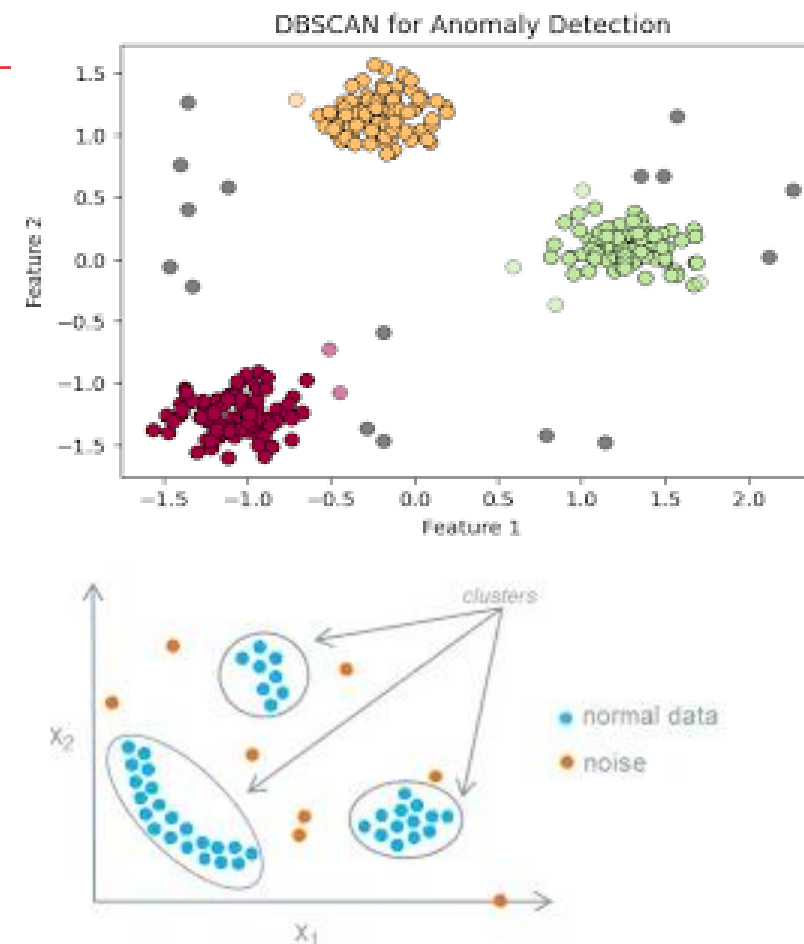
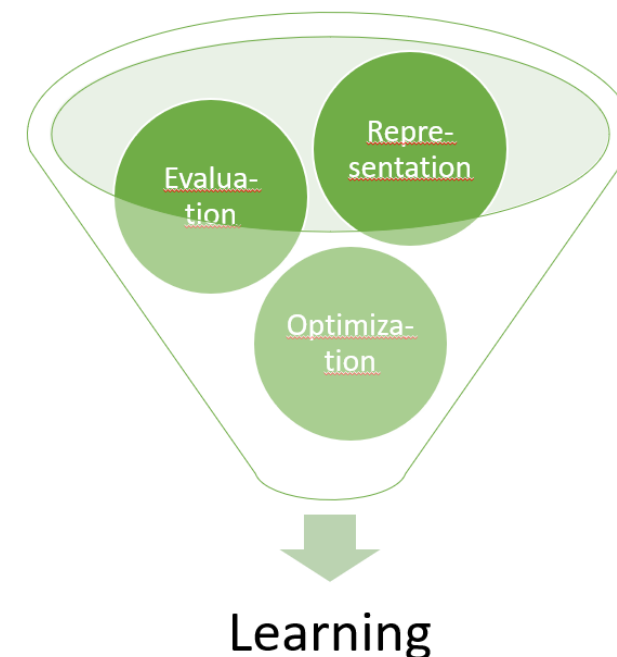


Fig 6: Anomaly Detection with K-Means Clustering

Key Elements of Machine Learning

- There are a good number of machine learning algorithms in use by data scientists today. In fact, some research indicates that there are perhaps tens of thousands. In addition, hundreds of new algorithms are put forward for use every year.
- Based on popular opinion, all machine learning algorithms today are made up of three components:
 - **Representation**: what the model looks like; how knowledge is represented.
 - **Evaluation**: how good models are differentiated; how programs are evaluated.
 - **Optimization**: the process for finding good models; how programs are generated.



Pedro Domingos, A Few Useful Things to Know About Machine Learning, CACM 55(10), 2012

Key Elements of Machine Learning

Representation
Instance-based
Hyperplane
Decision Tree
Set of Rules
Neural Networks
Graphical Models

Evaluation
Accuracy
Precision, Recall
Mean Avg Precision
Mean Average Error
Mean Square Error
K-L divergence
Likelihood

Optimization
Greedy Search
Branch and Bound
Gradient descent
Quasi-Newton
Linear Programming
Quadratic Programming

- Contoh: Anda mengklasifikasikan seseorang akan membeli komputer atau tidak, menggunakan model Multilayer Perceptron → Representation: neural networks, Evaluation: Accuracy, Optimization: gradient descent.

KA di Fasilkom UI

Kecerdasan Artifisial & Sains Data Dasar

- Intro to AI & DS
- Data preparation basics, dimensionality reduction (PCA)
- Clustering basics (K-means, hierarchical)
- Basic tree-based models (CART, random forest)
- Basic nearest neighbor classifier/regressor
- Simple probabilistic model (Naive Bayes)
- Simple linear models
- Basic neural networks
- Model evaluation basics (metrics, bias-variance tradeoff)
- Imbalanced classification

Pemelajaran Dalam (Deep Learning)

- Decision theory
- Optimization algorithm
- Linear models (linear regression, logistic regression, softmax regression, ridge & lasso regression)
- Deep fully-connected neural networks
- Deep convolutional neural networks
- Recurrent neural networks (vanilla RNN, GRU, LSTM)
- Attention mechanism & transformers

Pemelajaran Mesin (Machine Learning)

- Linear models, Maximum margin classification. Classification errors regularization. Logistic & Linear regression;
- Active learning. Kernels. Kernel regression; Support vector machine (SVM).
- Model selection. Model selection criteria;
- Description length. Feature selection;
- Combining classifiers. Boosting; Margin and complexity. Margin and generalization;
- Mixtures Model; Expectation maximization (EM) Algorithm. EM regularization;
- Markov models. Hidden Markov models (HMMs);
- Bayesian networks. Learning bayesian networks; Probabilistic inference.

Graf Pengetahuan (Knowledge Graph)

- Graph data model
- Persistent identifiers (URI)
- Resource Description Framework (RDF)
- Querying RDF data with SPARQL
- RDF semantics (RDFS)
- Linked data publishing
- Ontology & vocabularies
- Lightweight vocabulary: SKOS
- Ontology language OWL:
- RDF data validation with SHACL

Temu-balik Informasi (Information Retrieval)

- Intro to IR
- Indexing: inverted index, dictionary, external memory indexing, distributed indexing, dynamic indexing
- Phrase queries & proximity search
- Index construction, tokenization, normalization
- Index compression
- Ranked retrieval, Vector space model, BM 25
- Query processing, expansion
- Learning to rank, PageRank

Pengolahan Bahasa Manusia (Natural Language Processing)

- Regular expression
- Tokenization, stemming, lemmatization
- N-gram language model
- Vector semantics & embeddings
- Neural language models
- POS tagging, Named entity recognition
- Constituency & dependency parsing
- Logical representation of sentence meaning
- Coreference resolution
- Discourse coherence, Machine translation, Information extraction, question answering
- Chatbot & dialog systems

Penambangan Data (Data Mining)

- Data Mining Process; Statistical Limits on Data Mining;
- Pattern mining; Frequent Itemsets;
- Similarity & Distances; Cluster Analysis (Advanced Concept);
- Outlier Analysis;
- Text Mining; Human Annotation;
- Mining Discrete Sequences; Graph Mining;
- Social Network Analysis; Graph Representation

Computer Vision

- Computer vision tasks, human vision, perception, image understanding, color vision, color images, image acquisition, 2D to 3D transformation, digital image anatomy,
- Point operators, edge detection, subsampling dan super-resolution, morphological processing,
- Frequency domain, Discrete Fourier Transform, Wavelet Transform, Frequency Filtering,
- Monocular vs binocular, Stereo vision, photometric and multi-view stereo,
- Image Features, Feature selection and enhancement, Feature matching,
- RANSAC, image alignment, Template matching, SIFT,
- Convolutional neural networks,
- image classification and object detection in images,
- Image segmentation, image restoration, Image translation,
- Autoencoders, Generative Models, Image Synthesis, Image Generation,

Pemrosesan Bahasa Lisan (Spoken Language Processing)

- Pengenalan Pemrosesan Bahasa Lisan;
- Speech Theory; Fonetik Akustik;
- Automatic Speech Recognition (Speech-to-text);
- Speech Synthesis (Text-to-speech);
- Spoken Language Dialogue system

Robotika

- Robotika: permasalahan, tantangan dan aplikasi;
- Lokomosi pada robot berkaki dan beroda, UAV;
- Kinematika robot;
- Sensor pada robot; Ekstraksi fitur dan representasi fitur pada robot;
- Lokalisasi dan representasi peta pada robot;
- Path Planning; Aplikasi pendeteksi objek pada robot; Implementasi robot menggunakan perangkat lunak simulasi

Representasi Pengetahuan dan Penalaran (Knowledge Representation & Reasoning)

- First-Order Logic,
- representasi knowledge dan entailment, resolusi,
- production systems, structured descriptions,
- default reasoning, probabilistic reasoning,
- explanation dan diagnosis, reasoning about actions.

Pengelolaan Data Besar (Big Data Management)

- Introduction to Big Data and Data Analytics
- Introduction to HDFS, Big Data Ecosystem
- Classification and Regression
- Data Preprocessing for Analytics
- Clustering Techniques
- Apache Spark, Big Data Management and Maturity Model, NoSQL, Big Data Tech Presentation.

Pemrograman Logika (Logic Programming)

- Clause pada program logika: fakta, rule, query; Objek data pada Prolog;
- Semantik deklaratif program logika & logical entailment;
- Semantik prosedural program logika & resolusi;
- List, operator, dan aritmetika pada Prolog;
- Cut pada Prolog; Negasi pada Prolog;
- Operasi pada struktur data;
- Predikat dinamis dan pengumpulan solusi pada Prolog;
- Manipulasi term dan input/output pada Prolog;
- Difference list; Last call optimization dan accumulator

Inteligensi Bisnis

- Decision Support System and Business Intelligence
- Decision Making, Modelling, and Support
- Business Reporting and Visual Analytics
- Business Performance Management
- Data Warehouse, Business Intelligence Case Study
- Model-Based Decision: Multi-Criteria Optimization
- Data Mining for Business Intelligence

Analitika Media Sosial

- social media text analytics
- social network analysis
- location-based analysis
- social media user profiling



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